Learnable Path-Complex Features and Unified Path–Cellular Message Passing for Link Prediction

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Motivation

1-WL bounds plain GNNs [7]. Topological lifts escape it by message-passing over a richer object:

- **MPSN** [2]: biased to *cliques*.
- **CWN** [1]: biased to *induced cycles*.
- **PCN** [6]: simple paths, no substructure bias.

PCN leaves two problems on the table:

1. Symmetric scatter-sum init forces $A{\to}B{\to}C$ and $C{\to}B{\to}A$ to receive *identical* embeddings.
2. Lift is monolithic: paths *or* rings. Link prediction needs the closed-cycle signal that paths alone discard.

Our Contributions

1. **Position-aware learnable init.** $k{+}1$ position-specific linear maps per k-cell: strictly subsumes the scatter-sum and breaks orientation symmetry at $t{=}0$.
2. **Unified path and cellular complex.** Path- and ring-cells coexist at dim 2; canonicalised edges route messages between them for free.
3. **Theory (U-PWL).** U-PWL $\geq$ PWL *and* U-PWL $\geq$ CWL(k-IC); strictly $>$ 1-WL.
4. **Cora ablation, 10 seeds.** Each modification cuts seed variance; unified lift achieves the best mean test AUC.

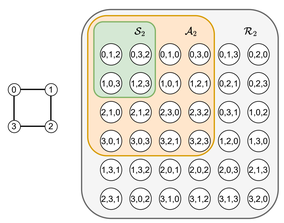
Big Takeaway

Unified path/ring cells with a learnable order-aware init: strictly greater expressivity and best Cora AUC at *zero* kernel cost.

- **Greater expressivity:** proved via U-PWL theory.
- **Lower variance:** -31% test-AUC std across 10 seeds.
- **Zero kernel cost:** same MP architecture as PCN.

Background: Path Complexes

A *simple k-path* is an ordered tuple $\sigma = (v_0, \dots, v_k)$ of distinct vertices with $\{v_i, v_{i+1}\} \in \mathcal{E}$; write $P_k(G)$ for the full set. The boundary alternates end-deletion, $\partial\, e_{i_0 \dots i_p} = \sum_q (-1)^q e_{i_0 \dots \hat{i}_q \dots i_p}$ [3], and PCN message-passes over the faces $\mathcal{B}(\sigma)$ together with the upper adjacencies $\mathcal{N}_\uparrow(\sigma)$.



$\mathcal{S}_2 \subset \mathcal{A}_2$: path complexes strictly enrich the 2-cell vocabulary.

The Deterministic-Init Problem

PCN initialises cell features by a symmetric aggregation, $h_\sigma^{(0)} = \bigoplus_{v \in \sigma} x_v$ with $\oplus \in \{\text{sum}, \text{mean}\}$ [6], which immediately collapses orientation: $h^{(0)}_{(A,B,C)} = h^{(0)}_{(C,B,A)}$. The damage is permanent, since later permutation-invariant aggregation also discards the signed boundary, so the source/target asymmetry that link decoders rely on is erased before message passing even starts.

This is not a quirk of the kernel choice: *any* symmetric $\oplus$ over the unordered vertex set of σ identifies a path with its reverse, collapsing the $(k{+}1)!$ orderings of a k-cell to a single representative at $t{=}0$. Downstream layers can never recover this lost orientation, capping PCN's expressivity strictly below what its boundary maps encode.

Method 1: Position-Aware Learnable Init

Replace the scatter-sum with $k{+}1$ position-specific maps:

$$h_\sigma^{(0)} = \sum_{i=0}^{k} W_i^{(k)} x_{v_i}$$

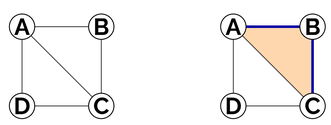
Each node contributes once per cell, so the asymptotic cost is unchanged. Sharing $W_i^{(k)} = W$ recovers a linear projection of the scatter-sum, so the new init strictly subsumes the deterministic one; for unequal weights, paths $\sigma \neq \sigma'$ with the same multi-set of endpoints receive *distinct* features, something no symmetric $\bigoplus$ can produce. The first vs. last endpoint encodes source vs. target, exactly the asymmetry link decoders already exploit. The price is $(k{+}1)\, d\, d_{\text{in}}$ parameters per k-cell ($d{=}64$ here); the init still runs in $\mathcal{O}(|P_k|\, d)$.

Method 2: Unified Path and Cellular Complex

Stack ring-cells *on top of* the path complex at dim 2:

$$C_2(G) = \underbrace{P_2(G)}_{\text{2-paths}} \cup \underbrace{R_L(G)}_{\text{induced cycles, } L \leq 7}$$

Path- and ring-cells have different boundary rules but share a single edge representation: a ring and a path on the same edge resolve to the *same* canonical edge cell, so upper-adjacency mixes ring $\leftrightarrow$ path messages for free.



(a) graph G **(b) $\mathcal{C}_2$: path *and* ring**

Figure 1. **Unified path and cellular complex.** On triangle ABC, the **ring-cell** (orange fill) and the **path-cell** $A{\to}B{\to}C$ (blue arrows) coexist and share edges AB, BC via canonicalised IDs, so $\mathcal{N}_\uparrow$ routes messages between them at zero kernel cost.

Why this targets link prediction

Common neighbours and short cycles drive link prediction (Adamic–Adar, SEAL). Path-cells expose only the *open* neighbourhoods of (u, v), whereas ring-cells expose the *closed* ones, supplying exactly the signal that path-only PCN misses.

Boundary rules differ; adjacency unifies

The path boundary is the signed alternating end-deletion $\partial(v_0, \dots, v_k) = \sum_q (-1)^q (v_0, \dots, \hat{v}_q, \dots, v_k)$, while the ring boundary collects the cycle's edges unsigned (the standard cellular ∂_2). Both reduce to the *same* canonical edge IDs, so a single $\mathcal{N}_\uparrow$ graph carries both kinds of message with no extra MP kernel. On Cora this adds +26% dim-2 cells; each triangle contributes 1 ring and 3 paths, so rings supply the triadic-closure signal that paths alone cannot encode.

Construction Recipe

As a one-off step, we enumerate $P_2(G)$ and $R_{\leq 7}(G)$ and canonicalise every edge by sorting its endpoints, so each 1-cell receives a *single* representation regardless of which 2-cell it bounds. Boundary and upper-adjacency tensors are built over $\mathcal{C}_2(G) = P_2 \cup R_{\leq 7}$; PCN message passing runs on top, position-aware init preserving orientation.

Worked Example: Symmetry Breaking

Take graph G: $a{-}b{-}c$ with one-hot features $x_a{=}e_1$, $x_b{=}e_2$, $x_c{=}e_3$. Compare reversed paths $\sigma{=}(a,b,c)$ and $\sigma'{=}(c,b,a)$.

Deterministic init collapses orientation:

$$h_\sigma^{(0)} = e_1{+}e_2{+}e_3 = h_{\sigma'}^{(0)}.$$

Learnable init with $W_0^{(2)}{=}I$, $W_1^{(2)}{=}\mathbf{0}$, $W_2^{(2)}{=}2I$:

$$h_\sigma^{(0)} = I\, e_1 + \mathbf{0}\, e_2 + 2I\, e_3 = e_1 + 2\, e_3,$$
$$h_{\sigma'}^{(0)} = I\, e_3 + \mathbf{0}\, e_2 + 2I\, e_1 = e_3 + 2\, e_1 \neq h_\sigma^{(0)}.$$

Orientation therefore **survives** initialization and flows through every later message-passing step.

Encoder to Decoder Pipeline

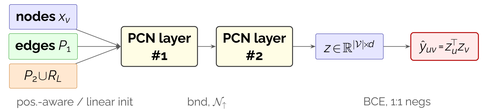


Figure 2. Cell features (dim-0,1,2) are initialised (linear for nodes, *position-aware* for paths and rings) and refined by two PCN message-passing layers; the node-level output feeds a dot-product link decoder trained with BCE.

Message Passing on the Unified Complex

Each layer aggregates boundaries and upper-adjacent cells:

$$m_{\mathcal{B}}(\sigma) = \sum_{\tau \in \mathcal{B}(\sigma)} M_{\mathcal{B}}(h_\tau^{(t)}), \quad m_\uparrow(\sigma) = \sum_{\tau \in \mathcal{N}_\uparrow(\sigma)} M_\uparrow(h_\tau^{(t)}),$$
$$h_\sigma^{(t+1)} = \text{UP}\big(h_\sigma^{(t)},\, m_{\mathcal{B}}(\sigma),\, m_\uparrow(\sigma)\big).$$

$M_{\mathcal{B}}$, $M_\uparrow$, UP: GIN-style MLPs [7].

Drop-in. The same sums silently mix **ring** and **path** messages with the *kernel unchanged*.

Theoretical Results

U-PWL: WL test on $\mathcal{C}(G)$

$$c_\sigma^{(t+1)} = \text{HASH}\left(c_\sigma^{(t)},\ \{\{c_\tau^{(t)}\}\}_{\mathcal{B}(\sigma)},\ \{\{c_\tau^{(t)}\}\}_{\mathcal{N}_\uparrow(\sigma)}\right)$$

- **Thm 1.** U-PWL $\geq$ PWL.
- **Thm 2.** U-PWL $\geq$ CWL(k-IC) at dim 2.
- **Cor.** U-PWL strictly $>$ 1-WL; not less than 3-WL.

Why: adding ring-cells preserves every PCN path-cell and adjacency, so U-PWL refines both PWL and CWL(k-IC) [2, 1, 6].

Cost & Complexity on Cora

Both modifications stay **cheap**: learnable init adds parameters only; the unified lift pays a one-off preprocessing cost. Per-epoch latency stays in the path-only budget. The MP kernel is identical across all four configs, so asymptotic compute is unchanged and lift cost amortises across seeds.

Config	Lift (s)	Epoch (ms)	Params
Path, Det.	22.0	28.5	428,544
Path, Learn.	22.0	37.5	703,680
Unified, Det.	**62.8**	**33.0**	**428,544**
Unified, Learn.	62.8	43.2	703,680

- **Learnable init:** +64% params, +32% epoch.
- **Unified lift:** +26% dim-2 cells; 2.9× preprocessing, *runs once*.
- **Asymptotic:** $\mathcal{O}(n \cdot b^{\max(k,L)})$ [6].

Cora Link Prediction: Setup & Results

- **Data:** Cora [5], $|\mathcal{V}|{=}2708$, $|\mathcal{E}|{=}5278$.
- **Split:** 85/5/10 [4], 1:1 negatives.
- **Model:** 2-layer encoder, hidden 64, dropout 0.3, dot decoder; $k_{\max}{=}2$, $L{=}7$.
- **Train:** Adam lr 0.01, 200 epochs, early-stop on val AUC. Mean±std over 10 seeds.

Complex	Init	Val AUC	Test AUC
Path	Det.	0.7344 ± 0.0274	0.7113 ± 0.0239
Path	Learn.	0.7354 ± 0.0230	0.7123 ± 0.0164
Unified	Det.	$\mathbf{0.7366 \pm 0.0270}$	$\mathbf{0.7132 \pm 0.0165}$
Unified	Learn.	0.7335 ± 0.0213	0.7111 ± 0.0175

Table 1. ROC-AUC, 10 seeds. **Learnable alone**: test std $0.0239 \to 0.0164$ (-31%). **Unified alone**: best mean test AUC. **Stacked**: partially redundant on Cora.

Per-Seed Variance

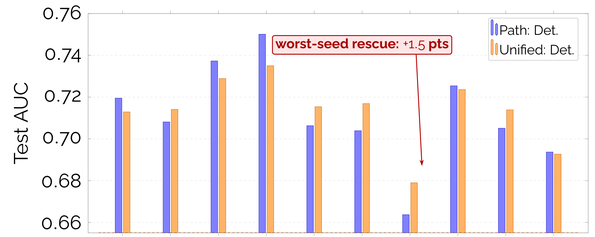


Figure 3. Per-seed test AUC; dashed lines mark the 10-seed means (Path 0.7113, Unified 0.7132). The unified lift lifts the worst-case seeds the most (seed 6: $0.664 \to 0.679$), consistent with the stronger triadic signal supplied by ring-cells.

Limitations & Future Work

Limitations. Cora gains are modest ($<1\%$ AUC); stacked modifications interact; closing the gap to SEAL / NBFNet requires larger benchmarks.

Future.

- Clique-cells alongside paths and rings.
- Larger benchmarks: OGBL-COLLAB / OGBL-PPA.
- Learnable gating to rebalance triangle over-weighting (each 3-cycle is one ring and three paths).

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